Fitness and Nutrition Tracker

DS242 – Advanced Data Science Programming
Saudi Electronic University

1. Introduction

This report provides a technical overview of the 'Fitness and Nutrition Tracker' project, developed using Python. This application's goal is to give users the ability to track their eating and exercise patterns via an easy-to-use, interactive interface. The application facilitates bulk import of entries, data visualization, profile maintenance, and meal and exercise logs. This project shows how Python programming may be used practically in data structures, file management, and simple data visualisation.

2. List of Methods :

• create_load_profile(username)
stores important information like height, weight, and fitness objectives while creating a new user profile or loading an existing one.

• change_profile(username)
makes sure a profile is accessible for interaction by using create_load_profile() to switch between user profiles.

• view_profile(username)
shows a history of all added entries as well as the user's physical characteristics and fitness objectives.

• edit_profile(username)
enables users to modify profile data, including height, weight, and desired level of fitness.

• add_entry(username)
enables users to manually add food or exercise logs by entering the date, category, type, quantity/duration, and calories.

• view_entries(username):
This function is helpful for monitoring progress over time since it displays all previous entries registered under the given user.

• import_csv_entries(username)
allows adding entries in bulk using a CSV file. Each row is read and verified by the system before the data is logged.

•import_xml_entries(username)

Though it parses structured XML files to create batch import entries, it is comparable to the CSV function.

• visualize_meal_distribution(username):

To examine eating patterns, this function creates a pie chart that summarises the various meal categories for a chosen month.

•visualize_calories_comparison(username)

gives users information about their energy balance by creating a bar chart that contrasts the number of calories expended during workouts with the number of calories ingested during meals.

3. Data Structures and File Integration

The application relies on essential Python data structures to manage user data and health logs effectively:

• **Dictionaries**: Every user's profile is saved as a dictionary including their body measurements, height, and fitness objectives. A dictionary including information such as date, category, calories, and type is also kept for every log entry (meal or workout).

```
entry = {
    'type': entry_type,
    'category': category,
    'duration_quantity': duration_quantity,
    'calories': calories,
    'date': entry_date
}
```

```
profiles[username] = {
    'profile': {
        'weight': None,
        'height': None,
        'fitness_goal': None
    },
    'entries': []
}
```

• **Lists**: A list of dictionaries is used to arrange the entries. This facilitates data aggregation, looping over logs, and applying visualisations.

Furthermore, the following formats can be used to import external data into the program:

• **CSV:** Users may upload enormous collections of historical information by using the integrated CSV module to parse tabular files.

• **XML:** Structured logs may be imported from prepared XML documents using Python's XML parsing modules.

4. Screenshots of the Outputs :

Sample output of running program and adding a new user, editing and viewing profile

```
              ###### Welcome to Fitness and Nutrition Tracker! ######


Please enter your profile name: FATIMA

 -> Profile does not exist... Creating new profile for user: FATIMA


 ==== Fitness & Nutrition Tracker Main Menu ====
1. Change profiles
2. View your profile
3. Edit your profile
4. View all entries
5. Add a new log entry interactively
6. Import entries from CSV
7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned


Select an option: 2


Your current profile stats:
Weight: None
Height: None
Goals: None


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Select an option: 3


Your current profile stats:
Weight: None
Height: None
Goals: None
```

```
Select an option: 1


Please enter profile name: Alkhnsaa

 -> Profile does not exist... Creating new profile for user: Alkhnsaa


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Select an option: 3


Your current profile stats:
Weight: None
Height: None
Goals: None

Enter new weight: 46
Enter new height: 158
Enter new fitness goal: gaun wight
Profile updated successfully.


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9. Visualize Calories Consume vs. Calories Burned
```

```
Please enter profile name: FarahAlhelal

 -> Profile does not exist... Creating new profile for user: FarahAlhelal


 ==== Fitness & Nutrition Tracker Main Menu ====
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Select an option: 3


Your current profile stats:
Weight: None
Height: None
Goals: None

Enter new weight: 47
Enter new height: 158
Enter new fitness goal: gain weight
Profile updated successfully.


 ==== Fitness & Nutrition Tracker Main Menu ====
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7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned


Select an option: 1


Please enter profile name: Alkhnsaa

 -> Profile does not exist... Creating new profile for user: Alkhnsaa
```

```
Please enter profile name: hanen

 -> Profile does not exist... Creating new profile for user: hanen


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Select an option: 3



Your current profile stats:
Weight: None
Height: None
Goals: None

Enter new weight: 155
Enter new height: 189
Enter new fitness goal: lose weight
Profile updated successfully.


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Select an option: 1



Please enter profile name: FarahAlhelal

 -> Profile does not exist... Creating new profile for user: FarahAlhelal
```

Adding a new entry interactively, changing profiles and viewing entries

```
Enter new weight: 45
Enter new height: 158
Enter new fitness goal: gain weight
Profile updated successfully.


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Select an option: 4


No entries recorded yet.


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4. View all entries
5. Add a new log entry interactively
6. Import entries from CSV
7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned


Select an option: 1


Please enter profile name: hanen

 -> Profile does not exist... Creating new profile for user: hanen


 ==== Fitness & Nutrition Tracker Main Menu ====
1. Change profiles
2. View your profile
3. Edit your profile
4. View all entries
5. Add a new log entry interactively
6. Import entries from CSV
```

```
Select an option: 1


Please enter profile name: FarahAlhelal

 -> Profile loaded for user: FarahAlhelal


 ==== Fitness & Nutrition Tracker Main Menu ====
1. Change profiles
2. View your profile
3. Edit your profile
4. View all entries
5. Add a new log entry interactively
6. Import entries from CSV
7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned


Select an option: 5


Enter type ('Workout' or 'Meal'): Meal
Enter date (YYYY-MM-DD): 2025-02-02
Predefined meal categories:  ['Breakfast', 'Lunch', 'Dinner', 'Snack']
Cjoose category (or type a new one): Breakfast
Enter quantity for meal: 550
Enter calories (burned for workout or consumed for meal): 300

 -> Meal entry added successfully.


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9. Visualize Calories Consume vs. Calories Burned


Select an option:
```

```
Please enter profile name: hanen

 -> Profile loaded for user: hanen


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6. Import entries from CSV
7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned


Select an option: 5


Enter type ('Workout' or 'Meal'):       Workout
Enter date (YYYY-MM-DD): 2005 8 27
Invalid date format. Use YYYY-MM-DD.


 ==== Fitness & Nutrition Tracker Main Menu ====
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Select an option: 5


Enter type ('Workout' or 'Meal'): Workout
Enter date (YYYY-MM-DD): 2025-03-04
Predefined workout categories:  ['Cardio', 'Strength', 'Flexibility']
Choose category (or type a new one): Strengh
Enter duration for workout: 30 min
Enter calories (burned for workout or consumed for meal): 300

 -> Workout entry added successfully.
```

```
Please enter profile name: FATIMA

 -> Profile loaded for user: FATIMA


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Select an option: 5



Enter type ('Workout' or 'Meal'): Workout
Enter date (YYYY-MM-DD): 2025-03-10
Predefined workout categories:  ['Cardio', 'Strength', 'Flexibility']
Choose category (or type a new one): Strenth
Enter duration for workout: 15 min
Enter calories (burned for workout or consumed for meal): 150

 -> Workout entry added successfully.


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4. View all entries
5. Add a new log entry interactively
6. Import entries from CSV
7. Import entries from XML
8. Visualize Meal Distribution
9. Visualize Calories Consume vs. Calories Burned



Select an option: 1
```

```
Please enter profile name: Alkhnsaa

 -> Profile loaded for user: Alkhnsaa


 ==== Fitness & Nutrition Tracker Main Menu ====
1. Change profiles
2. View your profile
3. Edit your profile
4. View all entries
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Select an option: 5



Enter type ('Workout' or 'Meal'): Meal
Enter date (YYYY-MM-DD): 2025-3-10
Predefined meal categories:  ['Breakfast', 'Lunch', 'Dinner', 'Snack']
Cjoose category (or type a new one): Lunch
Enter quantity for meal: 400
Enter calories (burned for workout or consumed for meal): 150

 -> Meal entry added successfully.
```

```
PROBLEMS (1)   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

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Select an option: 4



1.  [2025-02-02]      Meal     - Breakfast,       Duration/Quantity: 550,  Calories: 300
2.  [2025-02-01]      Workout  - Cardio,          Duration/Quantity: 30 minutes,   Calories: 250
3.  [2025-02-01]      Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 350
4.  [2025-02-02]      Workout  - Strength,        Duration/Quantity: 40 minutes,   Calories: 300
5.  [2025-02-02]      Meal     - Lunch,           Duration/Quantity: 1 plate,      Calories: 600
6.  [2025-02-03]      Workout  - Flexibility,             Duration/Quantity: 20 minutes,   Calories: 150
7.  [2025-02-03]      Meal     - Dinner,          Duration/Quantity: 1 serving,    Calories: 700
8.  [2025-02-04]      Workout  - Cardio,          Duration/Quantity: 35 minutes,   Calories: 270
9.  [2025-02-04]      Meal     - Snack,           Duration/Quantity: 1 piece,      Calories: 200
10.  [2025-02-05]     Workout  - Strength,        Duration/Quantity: 45 minutes,   Calories: 350
11.  [2025-02-05]     Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 360
12.  [2025-02-06]     Workout  - Flexibility,             Duration/Quantity: 25 minutes,   Calories: 180
13.  [2025-02-06]     Meal     - Lunch,           Duration/Quantity: 1 plate,      Calories: 650
14.  [2025-02-07]     Workout  - Cardio,          Duration/Quantity: 30 minutes,   Calories: 260
15.  [2025-02-07]     Meal     - Dinner,          Duration/Quantity: 1 serving,    Calories: 720
16.  [2025-02-08]     Workout  - Strength,        Duration/Quantity: 50 minutes,   Calories: 400
17.  [2025-02-08]     Meal     - Snack,           Duration/Quantity: 1 piece,      Calories: 220
18.  [2025-02-09]     Workout  - Flexibility,             Duration/Quantity: 20 minutes,   Calories: 170
19.  [2025-02-09]     Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 340
20.  [2025-02-10]     Workout  - Cardio,          Duration/Quantity: 40 minutes,   Calories: 280
21.  [2025-02-10]     Meal     - Lunch,           Duration/Quantity: 1 plate,      Calories: 630
22.  [2025-02-11]     Workout  - Strength,        Duration/Quantity: 35 minutes,   Calories: 320
23.  [2025-02-11]     Meal     - Dinner,          Duration/Quantity: 1 serving,    Calories: 710
24.  [2025-02-12]     Workout  - Flexibility,             Duration/Quantity: 30 minutes,   Calories: 190
25.  [2025-02-12]     Meal     - Snack,           Duration/Quantity: 1 piece,      Calories: 210
26.  [2025-02-13]     Workout  - Cardio,          Duration/Quantity: 45 minutes,   Calories: 290
27.  [2025-02-13]     Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 355
28.  [2025-02-14]     Workout  - Strength,        Duration/Quantity: 50 minutes,   Calories: 410
29.  [2025-02-14]     Meal     - Lunch,           Duration/Quantity: 1 plate,      Calories: 640
30.  [2025-02-15]     Workout  - Flexibility,             Duration/Quantity: 25 minutes,   Calories: 175
31.  [2025-02-15]     Meal     - Dinner,          Duration/Quantity: 1 serving,    Calories: 730
32.  [2025-02-16]     Workout  - Cardio,          Duration/Quantity: 35 minutes,   Calories: 265
33.  [2025-02-16]     Meal     - Snack,           Duration/Quantity: 1 piece,      Calories: 205
34.  [2025-02-17]     Workout  - Strength,        Duration/Quantity: 40 minutes,   Calories: 330
35.  [2025-02-17]     Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 345
36.  [2025-02-18]     Workout  - Flexibility,             Duration/Quantity: 30 minutes,   Calories: 195
37.  [2025-02-18]     Meal     - Lunch,           Duration/Quantity: 1 plate,      Calories: 620
38.  [2025-02-19]     Workout  - Cardio,          Duration/Quantity: 40 minutes,   Calories: 275
39.  [2025-02-19]     Meal     - Dinner,          Duration/Quantity: 1 serving,    Calories: 705
40.  [2025-02-20]     Workout  - Strength,        Duration/Quantity: 45 minutes,   Calories: 360
41.  [2025-02-20]     Meal     - Snack,           Duration/Quantity: 1 piece,      Calories: 215
42.  [2025-02-21]     Workout  - Flexibility,             Duration/Quantity: 20 minutes,   Calories: 165
43.  [2025-02-21]     Meal     - Breakfast,       Duration/Quantity: 1 bowl,       Calories: 350
44.  [2025-02-22]     Workout  - Cardio,          Duration/Quantity: 30 minutes,   Calories: 255
```
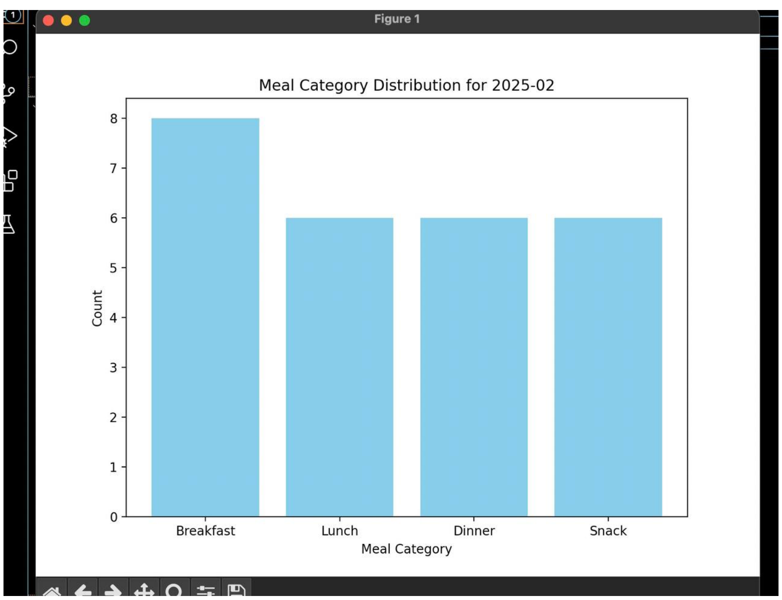

Visualizing meal distribution



5. Challenges and Solutions

Managing user input validation and integrating sophisticated features like predictive analysis were among the difficulties faced during the creation of the Fitness and Nutrition Tracker program. Some of the difficulties encountered and their corresponding fixes are listed below:

5.1 User Input Validation

- Challenge: Making sure that all user inputs, including dates, calorie counts, and entry kinds, are structured accurately to avoid mistakes.
- Solution: Every input method underwent a thorough validation check. When incorrect data is input, users are presented with unambiguous error warnings, preserving the quality of the data.

5.2 Data Structure and Integrity

- Challenge: Preserving consistency while importing data from external files or adding many items by hand.
- Solution: To identify and handle any discrepancies, the project makes use of structured dictionaries for profiles and entries in addition to error handling during the import procedure.

5.3 Accuracy of Visualisation

- Challenge: Accurately aggregating data when several entries take place on the same day to guarantee that visualisations show proper daily totals.

- Solution: Visualisation techniques compute the total number of calories burnt and eaten by aggregating data by date. This guarantees that accurate and significant insights are displayed in the pie and bar charts.

5.4 Handling File Imports

- challenge: Integrating data from XML and CSV files without corrupting the data

- Solution: To guarantee that only properly structured data is uploaded to the system, distinct procedures for CSV and XML imports were created, each with strong error handling and validation.

6. Conclusion

The project, which creates a fitness monitoring application, demonstrates Python programming abilities. It stores data in dictionaries and lists, imports external data via file operations, and provides user insights using visualisation approaches. The Fitness and Nutrition Tracker serves as an example of how programming may be used to meet practical health monitoring requirements.